

			$= 1260 \text{ N}$	A1
5	(a)		$V_R = 4.8 \quad (3.5)(0.80) \quad [\text{M1}]$ $= 2.0 \text{ V}$ $R = \frac{V}{I}$ $R = \frac{2.0}{0.80} = 2.5 \, \Omega \quad [\text{A1}]$	
5	(b)	(i)	B1 – Straight line graph that passes through the origin and (2.0, 0.8) within the accuracy of half a small square Allow ECF	
5	(b)	(ii)	From the graph, <u>current is proportional to potential difference</u> for an Ohmic resistor, (or <u>V to I ratio is constant</u>) [B1]	

			Since $I = nAqv$, when A, q and <u>number density n</u> are <u>constant</u>, <u>current is proportional to the drift velocity</u> and <u>drift velocity is proportional to the potential difference</u>. [B1]	
5	(b)	(iii)	$R_{\text{eff}} = \left(\frac{1}{5.0 + 1.5} + \frac{1}{3.5 + 2.5} \right)^{-1} \quad [\text{M1}]$ $= 3.1 \, \Omega$ $E = Pt = \frac{V^2}{R} t \quad [\text{C1}]$ $= \frac{4.8^2}{3.1} (3600)$ $= 2.7 \times 10^4 \, \text{J} \quad [\text{A1}]$ Alternative: $I_{\text{total}} = \frac{4.8}{5.0 + 1.5} + 0.80 \quad [\text{M1}]$ $= 1.54 \, \text{A}$ $E = Pt = VIt \quad [\text{C1}]$ $= 4.8 \times 1.54 \times 3600$ $= 2.7 \times 10^4 \, \text{J} \quad [\text{A1}]$ (Also accepted: $Q = It$, $E = VQ$)	
5	(b)	(iv)	Applying potential divider, $V_{1.5\Omega} = \left(\frac{1.5}{1.5 + 5.0} \right) 4.8 \quad [\text{M1}]$ $= 1.108 \, \text{V}$ $V_{\text{QP}} = 0.80 \times 3.5 - 1.108$ $= 1.7 \, \text{V} \quad [\text{A1}]$ 	